BI- DIRECTIONAL LSTM MODEL FOR CLASSIFICATION OF VEGETATION FROM SATELLITE TIME SERIES

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ABSTRACT

To further improve the classification accuracy of remote sensing time series data, in this paper, we propose a Bi-directional Long-Term and Short-Term Memory (denoted as BI-LSTM) based model for vegetation mapping and monitoring. The proposed model is applied to multi-temporal publicly available Sentinel-2A dataset with vegetation as the main theme. Experimental results have shown that the proposed approach has good performance in comparison with the state-of-the-art methods in term of accuracy, precision and recall. Moreover, it can efficiently use both past and future input features using the BI-LSTM component.

Index Terms—*land cover classification, vegetation modelling, recurrent neural network, long-short term memory, Bi-directional.*

1. INTRODUCTION

Recently, there has been significant growth and increasing amount of interest in time series classification. To monitor growing dynamics of vegetation, the multi-temporal remote sensing imageries is an important source of time series observations and analysis. Due to changes of reflective features over time, seasonality is considered as one of the most prominent characteristics of vegetation [1].

Several techniques have been proposed in the literature for multi-temporal optical remote sensing images classification [2]. In [3]-[4], authors employed various rule-based and supervised machine learning techniques tools to detect and to categorize the change of land cover [5].

Recently, deep learning Neural Networks techniques (NN) have been used to achieve promising results on a variety of land cover classification tasks including multi-temporal remote sensing images [5]-[6]. In [6], authors proposed a new Convolutional Neural Network (CNN) model to classify time series of remote sensing

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images and achieved a promising results. Similarly, Recurrent Neural Networks (RNN) have been widely used for multi-temporal classification [7]. In [8], authors proposed a Long Short-Term Memory (LSTM) deep neural networks to capture features and long-term dependencies of multi-temporal remote sensing images and achieved 91% of accuracy on the public sentinel-2A dataset [8]. However, the aforementioned approaches stills suffer and may fail to manage and to capture sufficient features. To address the limitation of feature extraction process and multi-temporal information modelling, in this paper we propose new model based on CNN and BI-LSTM to extract sufficient features for classification. Bi-directional LSTM is used in this study to help access to both backwarding and forwarding information extraction of multi-temporal data [9].

The proposed model is applied to Sentinel-2A dataset classification of time series with vegetation mapping as the main theme, and experimental results are evaluated with commonly used criteria.

The rest of this paper is structured as follows: In section 2, the proposed methodology is presented. In section 3 and 4, experiments and results are discussed. Finally, a conclusion and future work are drown in section 5.

2. PROPOSED APPROACH

Regarding the memory of RNN that is based on history information, LSTM are kind of neural network used for multi-temporal data classification. However, using single direction LSTM is inadequate and suffers from not gets access to both preceding and succeeding information at the same time. With the aim to learn the vegetation growth based on sequential observations via optical remote sensing images, we propose a BI-LSTM to enable the model to use both previous and future information via forward and backward states.

Figure 1 shows the architecture of the proposed model.

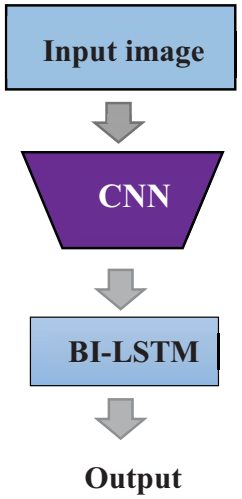


Fig. 1 The proposed model architecture.

3. EXPERIMENTAL SETTING

At this step, we address different hyper-parameters used to evaluate the proposed model through 10-fold cross-validation: The well-known Sentinel-2A remote sensing vegetation dataset are considered in this work (see Figure 2) [8]. We used 19 field classes from [8] as ground-truth information. The other class label is rejected if no field geometry was available. We used loss function classifier to control errors [10]-[11].

We enabled the dropout in training and disabled in evaluating the model and set dropout rate as 0.5. The considered metrics to evaluate the proposed approach are accuracy, Precision (P), Recall (R) and F-score as

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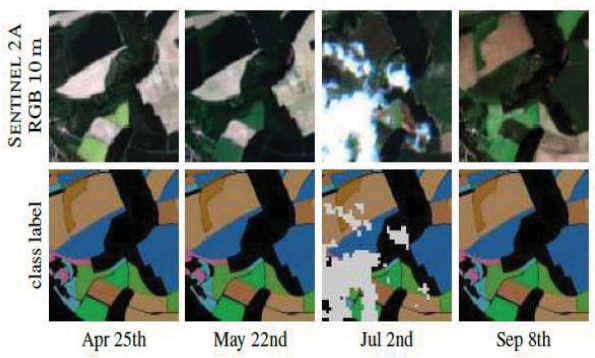


Fig. 2 Sentinel-2A images along the growth season 2016 [8].

described in [10]-[12]. The proposed model is tested and compared against the state of the art approaches.

4. RESULTS AND DISCUSSION

In this section, we discuss the details of the experimental results.

Table I. Illustration of multi-temporal vegetation classification accuracy.

Method	Accuracy	P	R	F-score
SVM [8]	31.1	31.4	31.1	31.1
CNN [8]	64.3	59.2	57.2	56.7
RNN [8]	72.9	77.3	73.0	74.0
LSTM [8]	74.3	78.4	74.5	75.3
CNN+LSTM model [13]	77.4	77.5	75.2	77.4
CNN+BI-LSTM model	**79.3**	**78.5**	**76.4**	**79.3**

Table I shows the results of the proposed approach. Comparing the results with state-of-the-art methods, the highest accuracy 79.3% is the best values, which is obtained by the proposed CNN+ BI-LSTM model with an improvement of 2%. Moreover, we further noted that the proposed model improves results more than 2% in terms of F-score (79.3%). Thus, it is evident that using BI-LSTM based on both forward and backward directions contribute to improve the vegetation time series classification accuracy.

Figure 3 shows the results of Kappa measure of the tested approaches using different networks. We can see that all the network architectures yield higher values in the first observations. While after day of year 100 (start period March-April), the kappa of (Bi-LSTM+CNN) increased constantly with length of sequence.

In addition, we further noted that after day of year 100 the temporal characteristic changes after the winter period with the vegetation period. This empirically confirm that using both past and future input features using the BI-LSTM component help to gets access to both preceding at the same time of multi-temporal vegetation, and outperforms the stat of the art multi-temporal remote sensing images.

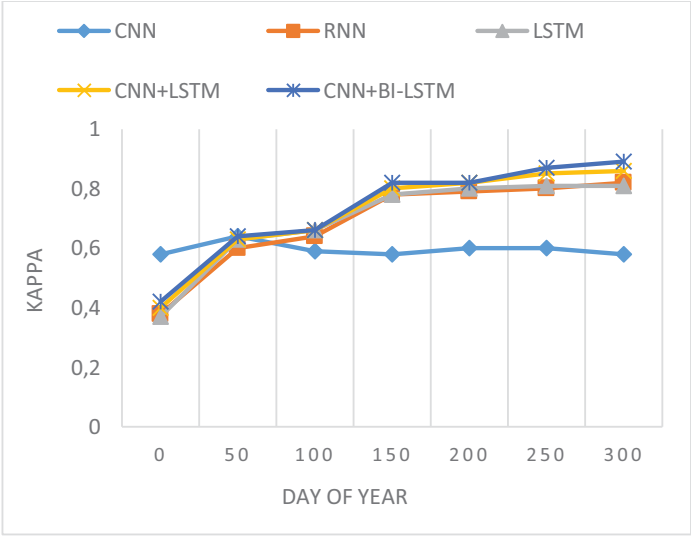


Fig. 3 The kappa measure of multi-temporal vegetation modeling using different networks.

5. CONCLUSION AND FUTURE WORK

In this paper, we presented a framework, based on CNN and BI-LSTM to learn multi-temporal vegetation of

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multispectral remote sensing images, which shows good accuracies in comparison with state-of-the-art methods. Our future work may consist in exploring other deep learning approaches to further improve the classification performance.

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